LEM-YOLO: Locality-Enhanced Mamba-YOLO Algorithm for Ship Detection in Remote Sensing Imagery

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***Abstract*—Ship detection in high-resolution optical remote sensing imagery holds significant value for applications. To address the issue of frequently missed small target detections, this study introduces a Mamba-YOLO architecture that utilizes global attention mechanisms to enhance the capabilities of small target detection. To further mitigate Mamba-YOLO's limitations in local feature modeling, we incorporate two modules, the Spatial-Channel Awareness Module (SCAM) and the Large Selective Kernel (LSK), between the backbone and neck networks. SCAM enhances discriminative feature representation through channel-wise attention, while LSK captures multi-scale contextual information via adaptive kernel selection. These localization-enhancing mechanisms considerably improve the recall of small targets. To tackle significant scale variations and arbitrary orientations of ships, we replace the detection head with an Oriented Bounding Box (OBB) design and integrate the ProbIoU loss function for accurately regressing rotationally oriented boxes. Extensive experiments on specialized datasets containing civilian vessels and warships demonstrate the effectiveness of our approach. The proposed model achieves a 3.3% improvement in mAP_{50} compared to the baseline Mamba-YOLO, confirming its superior performance across different ship categories. Index Terms: Ship detection, remote sensing, Mamba YOLO, SCAM, large selective kernel, ProbIoU loss.**

***Index Terms*—ship detection, remote sensing, Mamba YOLO, SCAM, large selective kernel, ProbIoU loss**

I. Introduction

In recent years, advances in high-resolution optical remote sensing imagery have significantly improved maritime surveillance capabilities. Automated vessel detection constitutes a pivotal component in the advancement of maritime governance and the operational efficacy of naval missions, as it furnishes near-instantaneous intelligence regarding both categorical classification and geospatial coordinates of seagoing craft. However, several challenges remain: ships often appear as small targets with weak features, leading to missed detections; complex backgrounds such as waves, clouds, and cluttered coastlines frequently cause false alarms; and the arbitrary orientations and aspect ratios of ships make traditional horizontal bounding boxes (HBB) inadequate, especially in crowded nearshore areas.

Local representations are hierarchically extracted through layered convolutional transformations within Convolutional Neural Networks (CNNs); however, long-range spatial dependencies remain intrinsically constrained by the limited receptive fields imposed by this architecture. [1].While modules such as the Path Ag-gregation Network (PANet) in YOLOv8 [2]enhance multi-scale feature fusion, they still struggle with significant scale variations, dense ship distributions, and small target detection in remote sensing imagery.

Recurrent neural networks (RNNs) achieve efficient sequence modeling through hidden state propagation [4]. A key advantage of RNNs is that inference requires only the present input vector and the antecedent hidden-state representation. Each state compresses historical information, thereby avoiding the need to recompute all prior states, as in Transformers, and enabling fast inference. However, this sequential nature prevents parallelized training, as computations must unfold temporally, significantly reducing training efficiency.

The Transformer architecture effectively models global context through self-attention [3] and accelerates training via parallel computation. Nevertheless, the quadratic complexity of matrix multiplications intrinsic to self-attention renders the attendant computational load prohibitive for high-resolution remote-sensing imagery, precluding deployment under resource-constrained conditions.

Each of these architectures offers distinct advantages, but also apparent limitations, motivating the need for a new model that balances representational capacity and computational efficiency. The recently proposed Mamba model combines the recurrent mechanism of RNNs with the representational capacity of state space models (SSMs) [4]. It supports input-dependent parameterization, parallel training, efficient inference, and dynamic feature selection, making it well-suited for detecting ships of various sizes in complex backgrounds with linear

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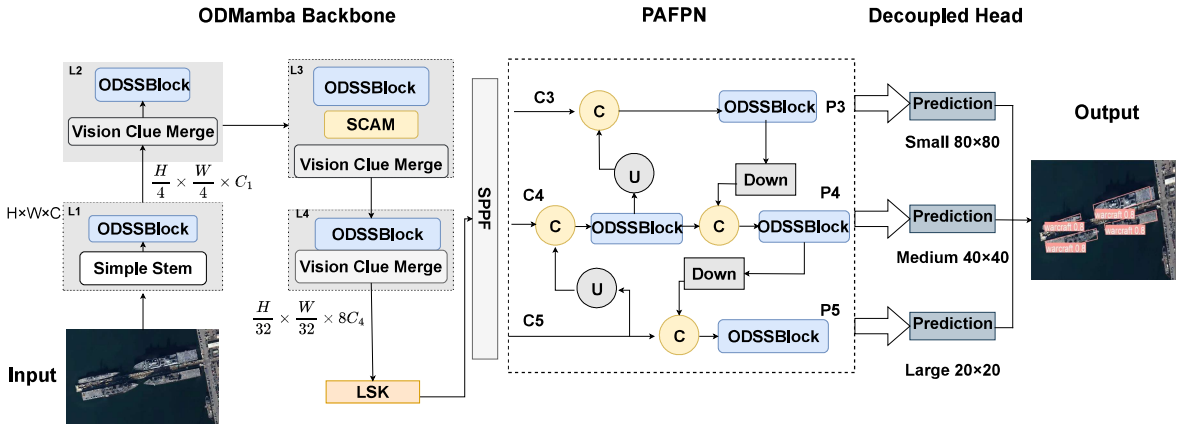


Figure 1: The proposed LEM-YOLO framework. LEM-YOLO utilizes the ODMamba as a backbone to extract multi-level features. The Spatial-Channel Awareness Module (SCAM) module is embedded at the L3 stage for spatial-channel attention enhancement, and Large Selective Kernel (LSK) blocks are applied after the L4 stage for multi-scale context capture. The ODSSBlocks are used for feature refinement and fusing, and the Vision Clue Merge blocks are for downsampling. The resulting features P3, P4, P5are output to the Decoupled Head for the prediction results.

computational complexity. Several recent works have already taken advantage of Mamba. For instance, Lu et al. [5]applied a dual-path Mamba design to video restoration, and Ding et al. [6] proposed XShuiMamba, a Mamba-based approach for efficient video summarization.

• We propose LEM-YOLO, an enhanced Mamba-YOLO framework that leverages Mamba for effective global context modeling, achieving a superior accuracy-efficiency trade-off for oriented ship detection in aerial images.

• The proposed LEM-YOLO framework integrates a spatial-channel awareness module (SCAM) and a large selective kernel module (LSK). The SCAM helps mitigate interference from complex backgrounds, augmenting the model's focus on small vessel targets through efficient channel and spatial attention. The LSK module aims to address scale variation by adaptively selecting optimal receptive fields, ensuring robust feature representation for both large and small ships.

• We incorporate a rotated detection head with Probabilistic Intersection-over-Union (ProbIoU) loss to solve the problem of dense arrangement and arbitrary orientation, significantly improving localization accuracy for oriented bounding boxes.

II. Related work

Aiming to tackle the difficulties posed by intricate backgrounds, target scale differences, and dense arrangements, ship detection has witnessed substantial advancements across various deep learning paradigms. This section systematically reviews the evolution of ship detection methodologies, with particular emphasis on their capabilities in addressing these persistent challenges.

Convolutional Neural Network (CNN)-based frameworks have demonstrated remarkable efficacy in ship detection. Detection methods based on directional perception, such as rotated bounding box prediction, significantly enhance the accuracy for multi-angle and densely arranged ships, overcoming the limitations of traditional horizontal bounding boxes [8], [9]. Faster R-CNN and its extended version Mask R-CNN have been widely adopted, with the latter providing instance segmentation masks for finer contour delineation [10], [13]. Liu and Stathaki enhanced the Faster R-CNN framework by integrating a semantic segmentation network that leverages high-level semantic cues to reduce false positives in challenging scenarios [12]. Liu et al. further proposed a coupled network incorporating a gated multi-layer feature extraction module to handle scale variations and a deformable occlusion-handling sub-network to address partial occlusions [11].

The YOLO series has gained widespread popularity in ship detection due to its exceptional balance between speed and accuracy. Various optimized solutions have been developed to address the specific challenges of maritime environments. LW-YOLO11 introduces a lightweight arbitrary-oriented detection mechanism specifically designed for multi-angle ship targets in complex sea conditions [29]. YOLO-RSA proposes a multiscale detection algorithm tailored for optical remote sensing images, effectively addressing scale variation issues in ship targets [30]. CSD-YOLO incorporates a deformable large kernel attention mechanism to enhance feature representation capabilities for ship detection [32]. Additionally, addressing the specific requirements of infrared imaging and small target detection, Xu et al. developed infrared target detection methods for complex sea background conditions [28], while Zhang et al. proposed detection techniques specifically designed for small targets on water surfaces in unmanned vessel scenarios [31].

More notably, foundation model approaches have opened new directions for remote sensing applications. Li et al. proposed SARATR-X, which aims to build a foundation model for SAR target recognition, demonstrating the potential of large-scale pre-trained models in specialized remote sensing tasks [34]. This direction aligns with the broader trend of self-supervised learning in remote sensing, as evidenced by research on joint-embedding predictive architectures for SAR ATR, showing that gradient prediction can produce more robust feature representations [36]. These modern approaches

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build upon fundamental research in feature representation, such as texture classification from random features [35], continuously advancing the theoretical foundations of remote sensing image analysis.

III. Method

A. Network Structure

As depicted in Figure. 1, the Mamba-YOLO network [7] incorporates the Selective State Space Model (S6) to achieve global feature modeling with linear computational complexity. The architecture comprises several key components: a Simple Stem that performs two stages of downsampling; a Vision Clue Merge mechanism that splits the feature map and concatenates segments along the channel dimension to reduce computational cost (FLOPs); and an SPPF module that reuses a single max-pooling kernel to generate multi-scale feature representations. The feature pyramid is constructed using a PAFPN-SSM structure, which employs three stacked ODSSBlocks to enhance global contextual awareness and refine multi-scale feature extraction, outputting feature maps P3, P4, P5. Finally, task-specific decoupled heads are used for independent classification and regression operations.

The Vision Clue Merge module leverages the CSPDarknet Focus module to split the feature map, resulting in four feature layers whose length and width are halved compared to the input image, and then stacks these four feature layers. Information on width and height dimensions is concentrated on the channel dimension. The 1 × 1 point convolution completely maintains the spatial resolution of the output feature map of the pre-stage SS2D module by eliminating the spatial dimension calculation of the convolution kernel, while ensuring that the high-frequency semantic information density does not decay, thereby avoiding the decay of traditional 3 × 3 convolution. The interference of the SS2D selective operation optimizes the downsampling process, retaining more visual cues.As the core module of Mamba YOLO, ODSSBlock primarily comprises three modules: LSBlock, RGBlock, and SS2D, which enable the network to learn deeper and richer feature representations. SS2D scans the four sequences obtained by Scan Merge, respectively, and uses the state space model Mamba to process the output of each sequence, thereby paying more attention to key information. It then merges the sequences from different directions to restore the feature map. The stable distribution is obtained by layer normalization. It improves the global context modeling ability of image features.

Building upon this foundation, we enhance the network by embedding the SCAM module after the p4/16 stage to refine semantic information through spatial-channel attention mechanisms, while introducing LSK blocks at the p5/32 level to improve multi-scale context perception using large selective kernels.

B. The Spatial–Channel Awareness Module

The Mamba-YOLO backbone extracts spatial features via the global state-space model, yet lacks explicit local-feature attention. Motivated by FFCA-YOLO [15], we insert a Spatial–Channel Awareness Module (SCAM) to remedy the above limitation. SCAM suppresses cluttered backgrounds and accentuates object-background distinctions.

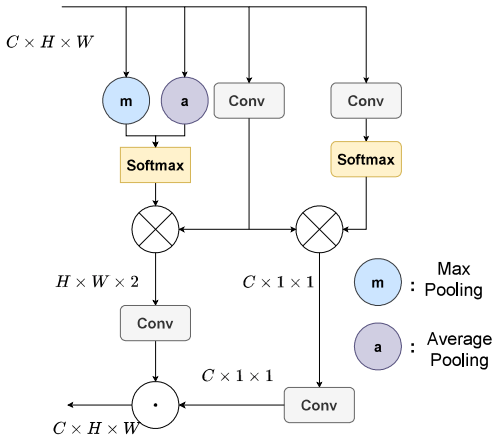


Figure 2: Structure of the spatial–channel awareness module.

As shown in Figure. 2, SCAM contains three branches. The channel attention branch employs dual-pooling (global average and max pooling) followed by nonlinear FC layers to generate a channel weight matrix $\mathbf{W_c}$, enabling dynamic channel selection. The channel transformation branch applies 1×1 convolution to reorganize channel features, reconstructing cross-channel correlations while preserving spatial resolution. The QK interaction branch efficiently captures long-range spatial dependencies by generating Query ($\mathbf{Q}$) and Key ($\mathbf{K}$) features via 1×1 convolutions and constructing a pixel-wise correlation map through matrix multiplication. The outputs of the three branches are fused via the Hadamard product:

$$\mathbf{Y} = \mathbf{B}_{\text{att}} \odot \mathbf{B}_{\text{trans}} \odot \mathbf{B}_{\text{QK}}$$

This element-wise multiplicative fusion retains individual branch patterns while amplifying jointly activated regions, forming a more discriminative feature representation.

C. The Large Selective Kernel Module

Conventional convolutional operations rely on preset, fixed kernel sizes, which lack the adaptability to the varying content of feature maps. To address this limitation, we introduce a dynamic alternative inspired by the Large Selective Kernel (LSK) module [17]. Our method employs the LSK to enable content-adaptive selection of receptive field sizes. This mechanism allows the network to more effectively capture multi-scale features by simultaneously attending to both local details and global contexts within a feature map.

The input feature $\mathbf{X}$ is processed through two branches. One branch employs two successive large-kernel convolutions to capture broad spatial contexts, yielding features $\mathbf{U}_1$ and $\mathbf{U}_2$. These features are concatenated channel-wise, after which average and max pooling are applied to generate spatial statistics. A subsequent convolution layer produces a spatial attention map $\mathbf{A}$ from these pooled features. The original features $\mathbf{U_1}$

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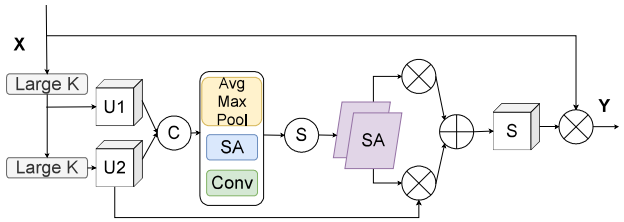


Figure 3: The LSK-Selection sub-block.

and $\mathbf{U_2}$ are then gated by $\mathbf{A}$ via Hadamard product, enhancing task-relevant regions. The resulting features are summed to form $\mathbf{S}$. The module's final output is obtained as:

$$\mathbf{Y} = \mathbf{X} \odot \mathbf{S}$$

This design allows the model to emphasize features from the most relevant spatial scales adaptively.

D. Rotating detection head and ProbIoU loss function

Remote-sensing ships exhibit arbitrary orientations and large aspect ratios. Horizontal bounding boxes thus introduce excessive background interference. We therefore replace the original Detect head with an Oriented Bounding Box (OBB) head.

The original rotating detection head employs the Complete IoU (CIoU) loss, which primarily penalizes horizontal overlaps while overlooking critical orientation information and exhibiting computational inefficiency in high-resolution scenarios. To address these limitations, we adopt the ProbIoU Loss. [24] This rotation-aware alternative models each oriented bounding box as a two-dimensional Gaussian distribution, resulting in more accurate and efficient regression.

In this probabilistic framework, a rotated box parameterized by $(x_{\text{center}}, y_{\text{center}}, w, h, \theta)$ is represented by its mean vector $\boldsymbol{\mu} = [x_{\text{center}}, y_{\text{center}}]$ and a covariance matrix $\boldsymbol{\Sigma}$ that encapsulates both orientation and scale information. The covariance matrix is derived from the box's geometry:

$$a' = \frac{w^2}{12}\cos^2\theta + \frac{h^2}{12}\sin^2\theta, \tag{1}$$

$$b' = \frac{w^2}{12}\sin^2\theta + \frac{h^2}{12}\cos^2\theta, \tag{2}$$

$$c' = \frac{w^2 - h^2}{12}\sin\theta\cos\theta, \tag{3}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} a' & c' \\ c' & b' \end{bmatrix}. \tag{4}$$

For two rotating boxes represented as Gaussian distributions $\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$, we compute their similarity using the Bhattacharyya distance:

$$D_{\text{B}} = \frac{1}{8}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^{\top}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) + \frac{1}{2}\ln\frac{|\boldsymbol{\Sigma}|}{\sqrt{|\boldsymbol{\Sigma}_1||\boldsymbol{\Sigma}_2|}}, \tag{5}$$

where $\boldsymbol{\Sigma} = \frac{\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2}{2}$.

The Hellinger similarity is then obtained as:

$$H = \sqrt{e^{-D_{\text{B}}}}. \tag{6}$$

Finally, the ProbIoU loss is defined as:

$$\mathcal{L}_{\text{ProbIoU}} = 1 - H. \tag{7}$$

This formulation jointly encodes angle, center position, scale, and spatial overlap in a unified probabilistic manner. The analytical derivation of covariance parameters ensures computational efficiency, making ProbIoU particularly suitable for real-time detection of rotated ships in high-resolution remote sensing imagery.

IV. Experimental Results

A. Implementation Details

We conduct extensive experiments on the HRSC 2016 dataset. The HRSC2016 dataset is a high-resolution remote sensing ship dataset for ship target detection. It contains images of two scenes featuring seaships and nearshore vessels. There are 1061 effectively labeled images. These images are partitioned into a training set, a validation set, and a test set, comprising 436, 181, and 444 images respectively.In the experiment, the image size is adjusted to 640x640 (pixels) on the premise of maintaining the aspect ratio . HRSC2016 comprises 1,061 high-resolution images with sea and nearshore ships, using standard 436/181/444 splits.

Experimental Setting: The experiments were implemented on Ubuntu 20.04 employing Python 3.11, CUDA 12.1, and PyTorch 2.3.0, using an NVIDIA GeForce RTX 3070 (8 GB of VRAM). We train for 300 epochs at an input size of 640×640, using SGD (momentum 0.937, weight decay 5×10^{-4}) with an initial learning rate of 0.01 and SiLU activation. The batch size is set to 8.

B. State-of-the-art Comparison

Table II presents a thorough comparison of our proposed method with state-of-the-art approaches on the HRSC2016 dataset. The evaluation metrics encompass mean Average Precision at IoU threshold 0.5 (mAP_{50}), Recall, Precision, and F1-score, providing a holistic assessment of detection performance. Additionally, we report model complexity through parameter count (Params) and inference speed through Frames Per Second (FPS) to evaluate computational efficiency.

Our method achieves competitive performance across all metrics, demonstrating superior balance between accuracy and efficiency. The results highlight the effectiveness of our proposed modules in enhancing feature representation while maintaining real-time detection capability.

Table I summarizes the quantitative results. Mamba YOLO-T attains 81.0% mAP_{50}, outperforming the compared YOLO variants. This performance gain can be attributed to the Mamba backbone's superior ability to model long-range dependencies within the image , which is crucial for capturing the contextual information around ships in vast maritime scenes. While YOLOv8-OBB employs rotated boxes, its convolutional

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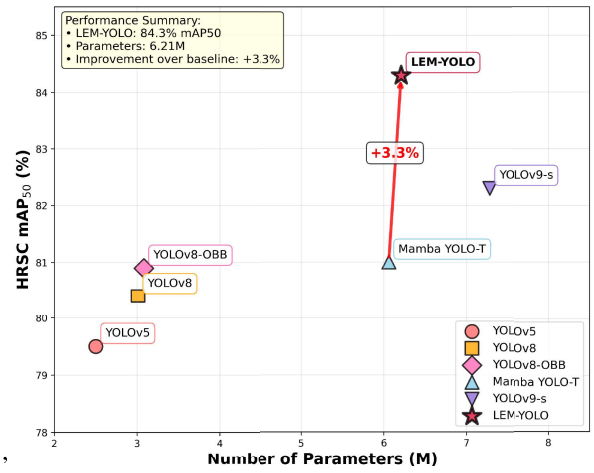


Figure 4: Comparison of mAP_{50} curves among different algorithms on the HRSC dataset. The results are evaluated in terms of mAP50 and the number of parameters (represented by the area of the circle).

TABLE I: Benchmark Algorithm Comparison

Model	mAP_{50} (%)	Recall	Precision	F1
YOLOv5	79.5	0.94	0.89	0.77
YOLOv8	80.4	0.92	0.91	0.76
YOLOv8-OBB	80.9	0.95	0.92	0.76
Mamba YOLO-T	**81.0**	0.91	**0.95**	0.76

TABLE II: Ablation Studies of Different Components on HRSC Dataset

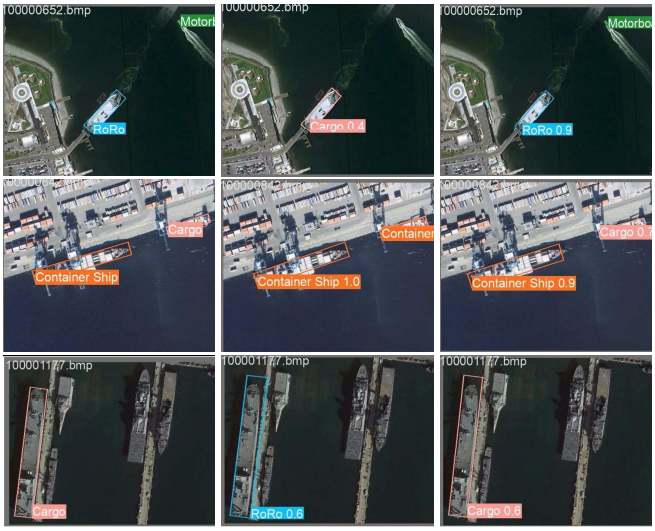
Configuration	**mAP_{50} (%)**	**Params (M)**	**FPS**
Baseline (Mamba YOLO-T + CIoU)	81.0	6.06	79.0
+ SCAM	82.8 (+1.8)	6.09	76.9
+ LSK	81.7 (+0.7)	6.09	72.3
+ ProbIoU	82.1 (+1.1)	6.06	72.4
All Components	**84.3 (+3.3)**	**6.21**	**70.2**

backbone lacks this global modeling capability, leading to slightly lower accuracy. The highest precision achieved by our baseline model also indicates its effectiveness in reducing false alarms among complex backgrounds.

C. Ablation Study

1) Effectiveness of SCAM: Table II shows that SCAM introduces only 0.03M parameters (+0.5%), yet achieves a significant improvement of 1.8% in mAP_{50}. The performance improvement is mainly due to the channel-space co-attention suppressing the shore clutter and wake artifacts, while maintaining precision. The spatial branch weighting enables small-sized ship features to obtain higher responses, significantly reducing missed detections.

2) Effectiveness of LSK: The LSK module demonstrates a performance improvement of mAP_{50} +0.7% on ship detection tasks, primarily attributed to its adaptive large kernel selection mechanism that effectively captures multi-scale characteristics of maritime targets. Ships typically exhibit varying sizes, diverse shapes, and complex backgrounds. By dynamically adjusting the convolution kernel's receptive field, LSK flexibly integrates local details with global context information, enhancing discrimination capability for ship structural characteristics while mitigating interference from surrounding waters.



(a) Annotated images (b) MambaYOLO (c)Ours

Figure 5: Qualitative comparison. (a) Ground truth, (b) Mamba-YOLO, (c) Our method.

3) Comparison of Loss Functions: Compared with CIoU in Table II, ProbIoU loss improves detection performance for rotated targets, achieving mAP_{50} enhancement of +1.1%. This improvement stems from ProbIoU's probabilistic modeling of the geometric distribution and directional characteristics of rotating ships, enabling more accurate characterization of bounding box uncertainty and overlap. The comprehensive integration of all proposed components demonstrates synergistic performance enhancement, achieving a substantial mAP_{50} improvement of 3.3% with only 0.15M additional parameters, while maintaining competitive inference speed.

D. Visual Verification

As shown in Figure. 5, our method correctly identifies the categories of ships that the Mamba YOLO misdetectsor misses. In the typical three-line scene, Mamba YOLO misclassifies Ro-Ro ships into cargo ships and missed motorboats, misclassifies small cargo ships into container ships, and misclassifies cargo ships into Ro-Ro ships. The method in this paper is all correctly identified and can detect smaller ships, and the missed detection rate is significantly reduced.

V. Conclusion

Optical ship images suffer from large scale variance, dense near-shore layouts and tiny targets. To capture long-range dependencies while preserving efficiency, we build a Mamba-YOLO detector and insert two plug-ins between the backbone and the neck: **SCAM** re-weights channels to suppress cluttered backgrounds, compensating for the missing channel attention in vanilla Mamba-YOLO; **LSK** adaptively selects kernel sizes to enlarge or shrink the receptive field on-the-fly, enabling

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robust multi-scale feature fusion. Finally, a rotated detection head supervised by ProbIoU loss is adopted to regress arbitrarily-oriented ships with higher precision and without extra inference cost.

Extensive experiments on HRSC2016 dataset demonstrate that our method LEM-YOLO outperforms YOLOv5, YOLOv8 and YOLOv8-OBB in both accuracy and speed, validating its effectiveness and efficiency. The proposed framework offers a comprehensive and efficient solution for detecting rotated objects in remote sensing imagery.

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